
RPW in 1D: Multi-Scale Heuristics vs. Grid Constructions

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Abstract

The convergence rate of empirical measures to their true distribution under the Wasserstein metric is subject to the curse of dimensionality. Recent work on the Robust Partial Wasserstein (RPW) distance introduces multi-scale proofs to establish tighter bounds, specifically improving the rate to $O(n^{-1/3})$ for W_2 in two dimensions. In this paper, we provide a rigorous analysis of these techniques specialized to the one-dimensional case ($d = 1$). We demonstrate that while the RPW multi-scale heuristic yields a rate of $O(n^{-2/5})$ for the refined metric, standard grid-based constructions yield strictly slower rates of $O(n^{-1/5})$ (1-grid) and $O(n^{-5/21})$ (2-grid). We conclude by contrasting these grid-based bounds with the optimal 1D sorting rate of $O(n^{-1/2})$, demonstrating that while RPW-inspired multi-scale methods are essential in higher dimensions, they are structurally suboptimal in one dimension where monotonic rearrangement is possible.

1. Introduction

Optimal Transport (OT) plays an increasingly central role in modern machine learning, with applications in generative modeling (Wasserstein GANs), domain adaptation, and distributionally robust optimization. A central theoretical question concerns the convergence of the empirical measure $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$ to a target distribution μ in the Wasserstein metric. This dependency implies that the sample complexity required to approximate the target distribution grows exponentially with the dimension d , rendering standard statistical estimation infeasible. Classical results show that $W_p(\mu_n, \mu)$ experiences curse of dimensionality: when $d > 2p$, the minimax rate deteriorates to $O(n^{-1/d})$

(Fournier & Guillin, 2015).¹ While alternative metrics such as Sliced Wasserstein (Bonneel et al., 2015) avoid this curse by projecting distributions onto 1D lines, determining sharp rates in the low-dimensional regime $d \leq 2p$ is substantially more delicate.

In two dimensions, the classical bound for W_2 is the $O(n^{-1/4})$ rate obtained through optimal matching arguments (Ajtai et al., 1984).² Recent work using the *Robust Partial Wasserstein* (RPW) metric introduced a new multi-scale trimming strategy that improves this to $O(n^{-1/3})$ (Raghvendra et al., 2024). This improvement arises from balancing the grid resolution with the fraction of mass discarded at each scale, a technique that becomes increasingly effective in higher dimensions.

This paper studies how these algebraic and geometric arguments behave in the one-dimensional setting ($d = 1$). Although the optimal rate for W_2 in 1D is known to be $O(n^{-1/2})$ via sorting, applying the RPW balancing principle to $d = 1$ produces a different exponent. We show that the multi-scale RPW heuristic yields a rate of $O(n^{-2/5})$, while explicit grid-based transport plans (1-grid and 2-grid) yield only $O(n^{-1/5})$ and $O(n^{-5/21})$, respectively. This highlights which parts of the RPW proof genuinely rely on high-dimensional geometry and which disappear in 1D.

Our contributions are as follows:

1. We specialize the RPW multi-scale cost-balancing argument to 1D deriving the exponent $O(n^{-2/5})$ for W_2 .
2. We construct explicit 1-grid and 2-grid couplings in 1D and show that they yield classical Wasserstein rates of $O(n^{-1/5})$ and $O(n^{-5/21})$, both strictly worse than the RPW heuristic and far from the optimal sorting rate.
3. We identify the precise mechanism responsible for the discrepancy between squared-cost balancing (W_2^2) and metric-level balancing (partial transport via $\Pi_{p,k}$), resolving a common source of confusion when deriving exponents in low-dimensional settings.

¹This follows from Fournier and Guillin, *On the rate of convergence in Wasserstein distance of the empirical measure*, PTRF (2015), Theorem 1, pp. 2–3.

²Ajtai, Komlós, and Tusnády, *On optimal matchings*, Combinatorica (1984), Theorem 1 in Section 2.

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2. Literature Survey

The analysis of empirical measure convergence has evolved from classical stochastic process theory to modern computational geometry and optimal transport. Our work sits at the intersection of these fields, specifically analyzing the behavior of recent robust metrics in lower-dimensional settings.

Foundations of Empirical Convergence. The fundamental study of how quickly an empirical distribution μ_n converges to its population limit μ dates back to the seminal work of [Dudley \(1969\)](#). In *The Speed of Mean Glivenko-Cantelli Convergence*, Dudley utilized chaining arguments and entropy numbers to bound the expected error of empirical processes. This work established that while convergence is guaranteed, the rate is heavily dependent on the "size" or complexity of the function class over which the distance is defined. This laid the theoretical groundwork for understanding why transport metrics, which can be viewed as supremum norms over Lipschitz functions (by Kantorovich-Rubinstein duality), suffer from dimension-dependent decay rates.

Sharp Rates for Standard Wasserstein. More recently, [Weed & Bach \(2019\)](#) provided a comprehensive characterization of these rates in *Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance*. They rigorously established that for the standard Wasserstein distance W_p , the convergence rate suffers from the curse of dimensionality, scaling as $n^{-1/d}$ when $d > 2p$. Crucially for our analysis, they highlight the regimes where "fast" rates are possible—specifically in one dimension ($d = 1$), where sorting yields a rate of $n^{-1/2}$. Their work serves as the baseline for standard grid-based constructions, confirming that without structural refinements, simple transport strategies cannot overcome the $n^{-1/d}$ barrier in high dimensions.

The Refined RPW Metric. To address the curse of dimensionality, [Raghvendra et al. \(2024\)](#) proposed *A New Robust Partial p -Wasserstein-Based Metric for Comparing Distributions*. They introduced the Refined p -Wasserstein (RPW) metric, denoted as $\Pi_{p,k}$, which incorporates negative Sobolev norms to penalize mass accumulation at different scales. By employing a recursive, multi-scale matching strategy (conceptually similar to Quadtree decompositions), they proved a convergence rate of $O(n^{-1/3})$ for $d = 2$, significantly improving upon the classical $O(n^{-1/4})$ rate implied by standard techniques.

In this work, we reverse-engineer the multi-scale mechanics proposed by [Raghvendra et al. \(2024\)](#). While their work focuses on the advantage of these heuristics in higher dimensions ($d \geq 2$), we apply their algebraic balancing techniques to the 1D case to explicitly contrast them against the concrete 1-grid and 2-grid constructions analyzed by [Weed &](#)

[Bach \(2019\)](#) and others.

3. Preliminaries

3.1. Wasserstein Distance

Let $\mathcal{P}(\Omega)$ be the set of probability measures on a compact domain $\Omega \subset \mathbb{R}^d$. Following the standard formulation in [Villani \(2009\)](#), the p -Wasserstein distance between $\mu, \nu \in \mathcal{P}(\Omega)$ is defined as:

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu, \nu)} \int_{\Omega \times \Omega} \|x - y\|^p d\gamma(x, y) \right)^{1/p} \quad (1)$$

where $\Gamma(\mu, \nu)$ is the set of couplings with marginals μ and ν .

3.2. The Robust Partial Wasserstein Metric

The RPW framework introduces a metric $\Pi_{p,k}$ designed to capture the multi-scale structure of transport. Conceptually, this metric penalizes both the local transport cost and the accumulation of mass fluctuations (negative Sobolev norms).

For our analysis, we rely on the core heuristic of the RPW bounds: the total error is controlled by the maximum of the local matching cost and the global fluctuation of the leftover mass.

4. The One-Dimensional Analysis ($d = 1$)

In one dimension, optimal transport enjoys a special structure via sorting, yielding the classical rate $W_2(\mu_n, \mu) = O(n^{-1/2})$ ([Weed & Bach, 2019](#)). In this section, we apply the multi-scale reasoning underlying the RPW metric ([Raghvendra et al., 2024](#)). Our goal is twofold: (i) to determine the convergence rate predicted by the RPW metric in one dimension, and (ii) to compare this rate with concrete grid-based transport constructions (1-grid and 2-grid) and the classical Wasserstein rate.

4.1. Method 1: The Multi-Scale RPW Heuristic

The RPW framework relies on balancing two quantities at a chosen spatial resolution h :

1. the *local matching cost* (within h -cells), and
2. the *global mass-fluctuation cost* (excess mass).

This yields a refined rate for the metric $\Pi_{p,k}$. We now carry out the full algebraic derivation derived from the probability of mass fluctuation.

STEP 1: GRID STRUCTURE AND STATISTICAL PROPERTIES

Consider a domain of dimension d (here $d = 1$) partitioned into a grid of size $h = n^{-\alpha}$. The volume of each cell is proportional to $n^{-\alpha d}$.

Let $X_1, \dots, X_n \sim \mu$ be i.i.d. samples. The number of samples N_i falling into a specific cell follows a Binomial distribution:

$$N_i \sim \text{Binomial}(n, p), \quad \text{where } p \approx n^{-\alpha d}.$$

The standard deviation of the count is given by $\sqrt{np(1-p)}$. For a fine grid, p is very small, allowing the approximation:

$$\text{SD}_{\text{count}} \approx \sqrt{np} = \sqrt{n \cdot n^{-\alpha d}} = n^{(1-\alpha d)/2}$$

STEP 2: EMPIRICAL MASS FLUCTUATION

The empirical mass in a cell is the count divided by n . Therefore, the fluctuation of the empirical mass (standard deviation of mass) is:

$$\text{SD}_{\text{mass}} = \frac{\text{SD}_{\text{count}}}{n} = \frac{n^{(1-\alpha d)/2}}{n} = n^{-\frac{1}{2} - \frac{\alpha d}{2}}$$

To find the *total leftover mass* (excess mass) across the entire grid, we multiply the typical fluctuation per cell by the number of intervals $m = n^{\alpha d}$:

$$\begin{aligned} \text{Total Excess Mass} &\approx (\# \text{ intervals}) \times (\text{fluctuation}) \\ &= n^{\alpha d} \times n^{-(1+\alpha d)/2} \\ &= n^{(\alpha d - 1)/2} \end{aligned}$$

STEP 3: BALANCING COST AND MASS

The RPW metric Π is controlled by the maximum of the local transport cost and the global excess mass:

$$\Pi \approx \max(\text{Cost}, \text{Mass}) = \max\left(n^{-p\alpha}, n^{(\alpha d - 1)/2}\right).$$

Here, the local cost scales as $h^p = n^{-p\alpha}$. To optimize the convergence rate, we equate the two exponents to find the optimal α :

$$-p\alpha = \frac{\alpha d - 1}{2} \implies p\alpha = \frac{1 - \alpha d}{2}$$

Rearranging for α :

$$2p\alpha = 1 - \alpha d \implies \alpha(2p + d) = 1 \implies \alpha = \frac{1}{2p + d}$$

STEP 4: FINAL RPW RATE FOR 1D

We now substitute the parameters for our specific 1D case ($d = 1$) and the 2-Wasserstein distance ($p = 2$):

$$\alpha = \frac{1}{2(2) + 1} = \frac{1}{5}$$

To find the overall rate, we substitute $\alpha = 1/5$ back into the cost expression:

$$\text{Overall Rate} = n^{-p\alpha} = n^{-2(1/5)} = n^{-2/5}$$

This confirms the heuristic rate calculation:

$$\min\left(2\alpha, \frac{1-\alpha}{2}\right) = \frac{2}{5} = 0.4$$

Theorem 4.1 (1D RPW Multi-Scale Rate). *For $p = 2$ and $d = 1$, the multi-scale RPW heuristic predicts*

$$\Pi_{2,k}(\mu_n, \mu) = \tilde{O}\left(n^{-2/5}\right). \quad (2)$$

Interpretation. The rate $n^{-2/5}$ arises from a competition between:

- *local* matching costs, which decrease as h^2 , and
- *global* mass fluctuations, which decay as $n^{-(1+\alpha)/2}$.

Balancing these two produces a strictly faster rate than any constant-depth grid construction (Sections 4.2–4.3), but slower than the optimal $n^{-1/2}$ rate achievable by exploiting monotonicity.

Comparison with 2D. It is worth noting that applying this same general formula $\alpha = \frac{1}{2p+d}$ to the 2D case ($d = 2, p = 2$) yields $\alpha = \frac{1}{6}$. This results in a convergence rate of $n^{-2(1/6)} = n^{-1/3}$, which matches the known bounds for 2D RPW give by (Raghvendra et al., 2024).

4.2. Method 2: The Concrete 1-Grid Construction

We now consider classical Wasserstein distance W_2 , which incurs a cost proportional to the *distance* transported. For a grid of spacing $n^{-\alpha}$:

- Local matching cost: $n^{-2\alpha}$.
- Leftover mass: $M \sim n^{(\alpha-1)/2}$.

Therefore,

$$W_2^2 \lesssim n^{-2\alpha} + n^{(\alpha-1)/2}.$$

Balancing:

$$-2\alpha = \frac{\alpha - 1}{2} \implies \alpha = \frac{1}{5}.$$

Thus:

$$W_2^2 \sim n^{-2/5} \implies W_2 \sim n^{-1/5}.$$

Proposition 4.2 (One-Grid Rate). *A one-grid transport plan in 1D yields the rate*

$$W_2(\mu_n, \mu) = O(n^{-1/5}).$$

4.3. Method 3: The Concrete 2-Grid Construction

Introducing a second grid allows the leftover mass from the fine scale to be partially corrected at a coarser resolution.

Let the fine grid scale be $n^{-\alpha_1}$ and the coarse grid scale be $n^{-\alpha_2}$. Using the same fluctuation estimates:

$$\gamma_1 \sim n^{-2\alpha_1}, \quad \gamma_2 \sim n^{(\alpha_1-1)/2} n^{-2\alpha_2}, \quad \gamma_3 \sim n^{(\alpha_2-1)/2}$$

Balancing $\gamma_1 = \gamma_2 = \gamma_3$ gives:

$$4\alpha_1 + \alpha_2 = 1, \quad 10\alpha_1 - 4\alpha_2 = 1,$$

whose solution is

$$\alpha_1 = \frac{5}{21}, \quad \alpha_2 = \frac{1}{21}.$$

Thus

$$W_2^2 \sim n^{-10/21} \implies W_2 \sim n^{-5/21}.$$

Proposition 4.3 (Two-Grid Rate). *A two-grid transport plan in 1D yields the rate*

$$W_2(\mu_n, \mu) = O(n^{-5/21}) \approx O(n^{-0.238}).$$

Comparison. The two-grid method improves upon the one-grid exponent $1/5 = 0.2$, but does not reach the RPW multi-scale exponent $2/5 = 0.4$, let alone the optimal sorting exponent $1/2 = 0.5$.

5. Comparison with 2D ($d = 2$)

To contextualize the 1D results, we review the RPW results for $p = 2, d = 2$. Using the multi-scale balancing formula $s = \frac{p}{2p+d}$:

$$s_{2D} = \frac{2}{2(2) + 2} = \frac{2}{6} = \frac{1}{3}. \quad (3)$$

The 2D rate is $O(n^{-1/3}) \approx n^{-0.333}$.

Lower dimensions converge faster due to smaller mass fluctuations relative to grid volume.

Table 1. Summary of Convergence Rates (Exponent s in n^{-s})

METHOD	$d = 1, p = 2$	$d = 2, p = 2$
OPTIMAL	$1/2$ (0.5)	$1/3$ (0.33)
CLASSICAL WASSERSTEIN	$1/2$ (0.5)	$1/4$ (0.2)
MULTI-SCALE HEURISTIC	$2/5$ (0.4)	$1/3$ (0.33)
2-GRID CONSTRUCTION	$5/21$ (0.238)	$3/10$ (0.3)
1-GRID CONSTRUCTION	$1/5$ (0.2)	$1/6$ (0.16)

6. The Case $p = \infty$

For the ∞ -Wasserstein distance,

$$W_\infty(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \|d(x, y)\|_{L^\infty(\gamma)}$$

the cost is determined by the *maximum* displacement rather than an average moment. When applying the RPW framework (Raghvendra et al., 2024), this leads to a modified scaling analysis compared to finite p .

6.1. RPW Heuristic for $p = \infty$

Let the grid scale be $h = n^{-\alpha}$. For $p = \infty$, transporting mass within a cell has cost proportional to h (not h^p). Thus:

$$\text{Local transport cost: } C_{\text{loc}} \sim h = n^{-\alpha}.$$

The leftover-mass fluctuation per cell, following the standard binomial deviation argument underlying empirical transport bounds (see, e.g., (Fournier & Guillin, 2015)), scales as

$$M \sim n^{(\alpha d - 1)/2},$$

so that the decay exponent is

$$s_{\text{mass}} = \frac{1 - \alpha d}{2}.$$

RPW requires balancing local cost and global mass fluctuation:

$$\alpha = \frac{1 - d\alpha}{2} \implies \alpha = \frac{1}{d + 2}.$$

Thus the RPW metric at $p = \infty$ satisfies the rate

$$\Pi_{\infty, k}(\mu_n, \mu) \lesssim n^{-1/(d+2)}.$$

Examples

$$d = 2 : \quad n^{-1/4} \qquad d = 1 : \quad n^{-1/3}$$

6.2. Comparison of Convergence Rates: RPW vs. Classical W_∞

To interpret the RPW bound derived, we contrast it with what is known about the empirical convergence of the classical bottleneck distance W_∞ . Unlike the $p < \infty$ Wasserstein distances, the metric W_∞ behaves qualitatively differently in statistical settings.

6.2.1. BIG-O RATE COMPARISON

The RPW framework produces the heuristic rate

$$\Pi_{\infty, k}(\mu_n, \mu) \lesssim n^{-1/(d+2)}.$$

This is a genuine convergence rate.

By contrast, *the classical W_∞ distance does not give a convergence rate in general, because it does not converge to 0 under empirical sampling.*

Thus the comparison table is:

Table 2. Comparison of Convergence Behavior ($p = \infty$)

DIMENSION	RPW (GRID BASED)	CLASSICAL W_∞ (EMPIRICAL BEHAVIOR)
$d = 1$	$O(n^{-1/3})$	NO CONVERGENCE TO 0
$d = 2$	$O(n^{-1/4})$	NO CONVERGENCE TO 0
d (GENERAL)	$O(n^{-1/(d+2)})$	NO CONVERGENCE TO 0

This makes clear that the RPW rate is meaningful, while classical W_∞ fails as a statistical metric.

6.2.2. WHY CLASSICAL W_∞ DOES NOT CONVERGE

Although $W_p(\mu_n, \mu) \rightarrow 0$ for every finite p , the case $p = \infty$ is fundamentally different. This failure manifests in two distinct ways.

1. Sensitivity to Outliers The definition is controlled by the single worst-matched point.

$$W_\infty(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \|d(x, y)\|_{L^\infty(\gamma)}$$

Thus,

$$W_\infty(\mu_n, \mu) \rightarrow \infty \quad \text{almost surely.}$$

2. Too strict The W_∞ metric is too strict because it measures the *maximum* distance any particle must travel. Since the empirical measure μ_n is discrete (a set of points) while the population μ is typically continuous, there are always gaps between samples.

Hence, the worst-case transport distance does not vanish, meaning

$$W_\infty(\mu_n, \mu) \not\rightarrow 0$$

almost surely, even for bounded distributions.

Consequence. In statistical settings, W_∞ is simply too strong to converge. RPW avoids this by truncating and redistributing mass across scales, yielding a well-defined, robust rate:

$$\Pi_{\infty, k}(\mu_n, \mu) \sim n^{-1/(d+2)}$$

7. Conclusion

This report has systematically analyzed the convergence rates for grid-based optimal transport constructions. We demonstrated that in 1D, the RPW multi-scale heuristic yields a rate of $O(n^{-2/5})$, which is superior to finite-grid constructions ($O(n^{-1/5})$ and $O(n^{-5/21})$) but inferior to the true optimal rate of $O(n^{-1/2})$ achieved by sorting.

This highlights that while multi-scale grid proofs are the state-of-the-art for $d \geq 2$, they are suboptimal techniques for $d = 1$, where the geometry allows for some rearrangement.

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